Consumption of Bing sweet cherries lowers circulating concentrations of inflammation markers in healthy men and women.

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The purpose of this study was to determine the effects of consuming sweet cherries on plasma lipids and markers of inflammation in healthy humans. Healthy men and women (n = 18) supplemented their diets with Bing sweet cherries (280 g/d) for 28 d. After a 12-h fast, blood samples were taken before the start of cherry consumption (study d 0 and 7), 14 and 28 d after the start of cherry supplementation (study d 21 and 35), and 28 d after the discontinuation (study d 64) of cherry consumption. After cherries were consumed for 28 d, circulating concentrations of C-reactive protein (CRP), regulated upon activation, normal T-cell expressed, and secreted (RANTES), and NO decreased by 25 ($P < 0.05$), 21 ($P < 0.05$), and 18% ($P = 0.07$) respectively. After the discontinuation of cherry consumption for 28 d (d 64), concentrations of RANTES continued to decrease ($P = 0.001$), whereas those of CRP and NO did not differ from either d 7 (pre-cherries) or d 35 (post-cherries). Plasma concentrations of IL-6 and its soluble receptor, intercellular adhesion molecule-1, and tissue inhibitor of metalloproteinases-2 did not change during the study. Cherry consumption did not affect the plasma concentrations of total-, HDL-, LDL-, and VLDL-cholesterol, triglycerides, subfractions of HDL, LDL, VLDL, and their particle sizes and numbers. It also did not affect fasting blood glucose or insulin concentrations or a number of other chemical and hematological variables. Results of the present study suggest a selective modulatory effect of sweet cherries on CRP, NO, and RANTES. Such anti-inflammatory effects may be beneficial for the management and prevention of inflammatory diseases.